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GEOG 465

4/28/26

Lab 4

These are my answers to the questions. At bottom of document I have placed most of the code I used throughout the assignment.

Answers

Question 1

How many total franchise restaurants are in the US?

The screenshot shows a PostgreSQL query editor interface. The query is as follows:

```
--SELECT * FROM pgis_ravindranathan.fastfoods;

-- // step 15
-- ALTER TABLE pgis_ravindranathan.fastfoods
-- ADD CONSTRAINT fk_fastfoods_franchise
-- FOREIGN KEY (franchise) REFERENCES pgis_ravindranathan.lu_franchises (franchise_code)
-- ON UPDATE CASCADE ON DELETE RESTRICT;

-- question 1
SELECT * FROM pgis_ravindranathan.lu_franchises;
```

The query results are displayed in a table with the following data:

franchise_code	franchise_name
b	Burger King
c	Carl's Jr
h	Hardee's
i	In-N-out
j	Jack in the Box
k	Kentucky Fried Chicken
m	McDonald's
p	Pizza Hut
t	Taco Bell
w	Wendy's

The interface also shows a status bar at the bottom indicating "Total rows: 10" and "Query complete 00:00:00.117".

Answer is 10 restaurants

Question 2

Provide a result table showing franchise name and total number of each franchise. Please order the result table in descending order of franchise total numbers. Take a screenshot image of the Data Output and provide it as an answer.

```
14 SELECT franchise_name, count(franchise) AS num_stores FROM pgis_ravindranathan.fastfoods
15 JOIN pgis_ravindranathan.lu_franchises ON franchise = franchise_code
16 GROUP BY franchise_name;
```

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1 of 1

	franchise_name character varying (100)	num_stores bigint
1	Hardee's	1802
2	in-N-out	203
3	Jack in the Box	2057
4	Pizza Hut	6214
5	Wendy's	5971
6	McDonal's	13745
7	Carl's Jr	987
8	Taco Bell	6164
9	Burger King	7435
10	Kentucky Fried Chicken	5424

I realized afterwards I forgot to order this output correctly, so here is a later screenshot:

Query Query History

```
1 SELECT franchise_name, count(franchise) AS num_stores FROM pgis_ravindranathan.fastfoods
2 JOIN pgis_ravindranathan.lu_franchises ON franchise = franchise_code
3 GROUP BY franchise_name
4 ORDER BY num_stores DESC;
```

Data Output Messages Notifications

Showing rows: 1 to 14 Page No: 1

	franchise_name character varying (100)	num_stores bigint
1	McDonal's	13745
2	Burger King	7435
3	Pizza Hut	6214
4	Taco Bell	6164
5	Wendy's	5971
6	Kentucky Fried Chicken	5424
7	Jack in the Box	2057
8	Hardee's	1802
9	Carl's Jr	987
10	in-N-out	203
11	MOD Pizza	1
12	Habit Burger & Grill	1
13	Chipotle	1
14	Panda Express	1

Total rows: 14 Query complete 00:00:00.256

Question 3

Q3. Please explain the below SQL codes in a paragraph and provide Data Output table image (0.5 pts)

Codes:

```
-- // 1.
```

```
-- SELECT ST_BUFFER(geom, 1609, 'endcap=round join=round ') as geom
```

```
-- FROM pgis_ravindranathan.i5seattle_p;
```

```
-- SELECT ST_UNION(ST_BUFFER(geom,1609,'endcap=round join=round')) as geom
```

```
-- FROM pgis_ravindranathan.i5seattle_p;
```

```
1 -- // 1.
2 SELECT ST_BUFFER(geom, 1609, 'endcap=round join=round ') as geom
3 FROM pgis_ravindranathan.i5seattle_p;
4
5 -- SELECT ST_UNION(ST_BUFFER(geom,1609,'endcap=round join=round')) as geom
6 -- FROM pgis_ravindranathan.i5seattle_p;
7
8 -- SELECT DISTINCT ff.ff_id, fr.franchise_name, ff.geom
9 -- FROM pgis_ravindranathan.fastfoods AS ff
10 -- JOIN pgis_ravindranathan.lu_franchises AS fr ON ff.franchise = fr.franchise_code
```

geom	geometry
01030000207308000001000000230000000272A0D02739C114482B8F1D31F418558D38A3E2839C172FAA5BC27D81F412B05EAC4F2839C13E0FFE7E13D01F419E446B8	
010300002073080000010000002C00000063A79BF481D39C1532B9F5E193F20414C8BD4F7CA1C39C1C04E9432FE4220414B3F51DFA51C39C1A08D7F5C2A4204186A900F	
010300002073080000010000002900000032FBC03E1939C14DE7214A0A221F41A930F6A1BE1939C1805568839201F419819ED007D1939C1D4CE01BB021D1F41F36F484	
010300002073080000010000002568BD01572139C1DAB5C7D8B8152041545279EBDE2039C10AD73481FF17204111E207822F2039C180B95CCB081A204176FF195	
010300002073080000010000002E00000056A84D6EE2739C18FEBF8E4F08D1F4124E6ABE1F42739C1E147A99FD8D1F416CEB9485042939C14A1C2A967D901F4195A2A61	
010300002073080000010000002F000000A848D4930C2339C16056C35920181F418C4903250D2339C1A024D2BF26181F41D75A7B0BE2239C18B3C84E95181F415C8FCB	
010300002073080000010000003900000075D87C8BA42639C185A1CE432AF91F41B9A2F25AE2639C15B0339E781F91F41DCE50FD1F42639C1D4C0EAB54FFE1F4173F28CE	
0103000020730800000100000026000000840FEA19992339C139AD89FC030F2041468B92E8BD239C1C9B9840B4E0F20416809859E172239C118A0578B3D122041D79B21E1	
010300002073080000010000002B0000005EB51788B41739C1FA6D51540C5C204168B52E14A51739C1F331AEEA7F5D2041410A6ACF4D1739C1F7C01B21DE5F2041C7A0D7	
01030000207308000001000000300000006A1B54C9B02839C14AD5F186ABDF1F41D15E7BCBD12839C1FF8E78B033E21F41C83BADF3D52839C1E1EDE68B1AE31F413351E5	
010300002073080000010000002300000013683961391C39C1168A21411FE41F4146CBFAF831C39C183D003ED310F14105324339681C39C10D92590D57DA1F4154AF380	
010300002073080000010000002C0000008C952C47601039C14839A89137412041C3EBF2D0D91039C1FE1D42BCBA4020413FC1F6B8C1039C159CFF7707E402041800FF2i	
010300002073080000010000002C0000002D61340F032039C176CEBBF5CE1E204110847126F41F39C121F57E52291F20410E24D22C4B1F39C1782F5CBFF8222041163E9C1A	

```
1 -- // 1.
2 -- SELECT ST_BUFFER(geom, 1609, 'endcap=round join=round ') as geom
3 -- FROM pgis_ravindranathan.i5seattle_p;
4
5 SELECT ST_UNION(ST_BUFFER(geom,1609,'endcap=round join=round')) as geom
6 FROM pgis_ravindranathan.i5seattle_p;
7
8 -- SELECT DISTINCT ff.ff_id, fr.franchise_name, ff.geom
9 -- FROM pgis_ravindranathan.fastfoods AS ff
10 -- JOIN pgis_ravindranathan.lu_franchises AS fr ON ff.franchise = fr.franchise_code
```

geom	geometry
01030000207308000001000000F80200009EE497AA462039C1ADDE0698F6A91E4194075267E22139C1FAS34ADAD9EA61E41D119EE6E3B2239C1AE96860B0CA41E41C396B96A6	

```
-- // 2.

-- SELECT DISTINCT ff.ff_id, fr.franchise_name, ff.geom

-- FROM pgis_ravindranathan.fastfoods AS ff

-- JOIN pgis_ravindranathan.lu_franchises AS fr ON ff.franchise = fr.franchise_code

-- JOIN pgis_ravindranathan.i5seattle_p AS i5 ON ST_DWITHIN(ff.geom, i5.geom, 1609*1)

-- ORDER BY ff.ff_id;

-- //Creating the ff View

-- CREATE VIEW pgis_ravindranathan.ffwithin1 AS

-- SELECT DISTINCT ff.ff_id, fr.franchise_name, ff.geom

-- FROM pgis_ravindranathan.fastfoods AS ff

-- JOIN pgis_ravindranathan.lu_franchises AS fr ON ff.franchise = fr.franchise_code

-- JOIN pgis_ravindranathan.i5seattle_p AS i5 ON ST_DWITHIN(ff.geom, i5.geom, 1609*1)

-- ORDER BY ff.ff_id;
```

The screenshot shows a database management tool interface. On the left, a sidebar lists various database objects: Publications, Schemas (2), pgis_ravindranathan (with sub-items like Aggregates, Collations, Domains, FTS Configurations, FTS Dictionaries, FTS Parsers, FTS Templates, Foreign Tables, Functions, Materialized Views, Operators, Procedures, Sequences, Tables (4), fastfoods, i5seattle_p, kingcounty, lu_franchises, Trigger Functions, Types, and Views (1)), and public. The main window displays a SQL query and its results. The query is the same as the one in the previous block. The results are shown in a table with columns ff_id, franchise_name, and geom. The table contains 10 rows of data, showing fastfood entries for Burger King and Pizza Hut franchises. The status bar at the bottom indicates 'Total rows: 54' and 'Query complete 00:00:00.175'.

ff_id	franchise_name	geom
1	Burger King	01010000207308000086C0ACAE1A1839C1ED2E801BCE562...
2	Burger King	010100002073080000601200DDE1D39C14EF80C3001B51...
3	Burger King	010100002073080000FD4C7F11300E39C1E6C909E465820...
4	Burger King	0101000020730800009778A8301F2039C140CF747013CA1E...
5	Burger King	0101000020730800007349DE81302639C102FB9A1297AA1F...
6	Burger King	010100002073080000AA5A2B7281F39C1396BA05D01E91...
7	Burger King	01010000207308000012EC6296A12239C17E836D5BB8901F...
8	Pizza Hut	0101000020730800004E0F0BA8490E39C1511B4570F05720...
9	Pizza Hut	0101000020730800000A1E18C2B72639C1F5EF896194F01E...
10	Pizza Hut	010100002073080000AD4B5BF061539C11E8A5F52B82720...

```
-- // 3.

SELECT fr.franchise_name, COUNT(DISTINCT ff.ff_id) AS tot

FROM pgis_ravindranathan.fastfoods AS ff

JOIN pgis_ravindranathan.lu_franchises AS fr ON ff.franchise = fr.franchise_code
```

```
JOIN pgis_ravindranathan.i5seattle_p AS i5 ON ST_DWITHIN(ff.geom, i5.geom, 1609*1)
```

```
GROUP BY fr.franchise_name ORDER BY tot DESC;
```

Output table:

```
21 -- // 3.
22 SELECT fr.franchise_name, COUNT(DISTINCT ff.ff_id) AS tot
23 FROM pgis_ravindranathan.fastfoods AS ff
24 JOIN pgis_ravindranathan.lu_franchises AS fr ON ff.franchise = fr.franchise_code
25 JOIN pgis_ravindranathan.i5seattle_p AS i5 ON ST_DWITHIN(ff.geom, i5.geom, 1609*1)
26 GROUP BY fr.franchise_name ORDER BY tot DESC;
27
```

Data Output Messages Notifications		
Showing rows: 1 to 7 Page No: 1		
	franchise_name character varying (100)	tot bigint
1	McDonald's	17
2	Jack in the Box	11
3	Burger King	7
4	Taco Bell	7
5	Kentucky Fried Chicken	4
6	Pizza Hut	4
7	Wendy's	4

The code under 1. creates a Buffer around the i5 layer. The first code chunk produces 190 buffers for the 190 segments of i5seattle_p table. The second code chunk condenses these buffers into one continuous buffer for the entire geometry of I5.

The code under 2. searches for restaurants within 1 mile of I5 (on the 1609 SRID). It joins the franchise and I5 information onto the fastfood-restaurants table, and then selects locations by geometry within the specified distance from I5. The entries are ranked by their id. The second code chunk turns this table query into a View for later access.

The code under 3. returns the number of restaurants per franchise within a mile of I5. It uses the same Joins and conditions as before, grouping by the franchise name and returning both name and count per franchise.

Q4. Get a center point at King County in WKT format

```

29
30 -- // 4.
31 -- SELECT "COUNTY", ST_CENTROID(geom) AS centroid FROM pgis_ravindranathan.kingcounty;
32
33 SELECT "COUNTY", ST_AsText(ST_CENTROID(geom)) AS centroid
34 FROM pgis_ravindranathan.kingcounty;
35

```

Data Output		Messages	Notifications
	COUNTY character varying (80)	centroid text	
1	KING	POINT(-1612831.6295027244 498633.490371453...	

Think the output from this code is the same as what Q4 asks for.

Q5. Find franchises within King County boundary. List names without overlap. To solve the question, it is necessary to do two joining process. One is joining lu_franchises and fastfoods to see franchise name in outcome and another is spatial joining fastfoods and kingcounty to find spatial relationship between them. Fill out the underlined space to complete SQL codes. The data output will look like the table in below. (0.5 pts)

```

32
33 -- SELECT "COUNTY", ST_AsText(ST_CENTROID(geom)) AS centroid
34 -- FROM pgis_ravindranathan.kingcounty;
35
36 -- // Q5
37 SELECT DISTINCT fr.franchise_name FROM pgis_ravindranathan.lu_franchises fr
38 JOIN pgis_ravindranathan.fastfoods ff ON ff.franchise = fr.franchise_code
39 JOIN pgis_ravindranathan.kingcounty kc ON ST_CONTAINS (kc.geom, ff.geom)
40
41 -- SELECT DISTINCT franchise_name FROM pgis_ravindranathan.lu_franchises

```

Data Output		Messages	Notifications
	franchise_name character varying (100)		
1	Jack in the Box		
2	Pizza Hut		
3	Wendy's		
4	McDonal's		
5	Taco Bell		
6	Burger King		
7	Kentucky Fried Chicken		

Compared to all the franchise names in lu_franchises, 3 restaurants were removed: Hardee's, in-N-out, and Carl's Jr.

I haven't seen stores from those 3 chains in King County, so I think the query worked correctly.

Q6. Where is the closest franchise restaurant from the King county center point? Run the code below and provide three names of the closest franchises. (0.5 pts)

```

48  -- // Q6
49  SELECT fr.franchise_name, ST_distance(ff.geom,
50      ST_centroid(pgis_ravindranathan.kingcounty.geom)) AS Distance
51      FROM pgis_ravindranathan.lu_franchises fr
52      JOIN pgis_ravindranathan.fastfoods ff ON ff.franchise = fr.franchise_code,
53      pgis_ravindranathan.kingcounty
54      ORDER BY Distance ASC; -- LIMIT 3
55

```

Data Output		
	franchise_name character varying (100)	distance double precision
1	McDonal's	260.28875412927
2	Burger King	266.2702786460578
3	Taco Bell	19398.802631640072
4	Kentucky Fried Chicken	19398.802631640072
5	Kentucky Fried Chicken	19537.363023882433
6	Taco Bell	19537.363023882433
7	McDonal's	19727.25206450137
8	Jack in the Box	19832.813232519507
9	McDonal's	20503.41400301334
10	Burger King	20786.23858829689
11	McDonal's	21254.9886075297
12	McDonal's	21959.973169353783
13	Jack in the Box	22657.382579879424
14	McDonal's	22708.417007291177
15	Jack in the Box	26628.892315177724

Total rows: 50002 Query complete 00:00:27.131

```

48 -- // Q6
49 SELECT fr.franchise_name, ST_distance(ff.geom,
50      ST_centroid(pgis_ravindranathan.kingcounty.geom)) AS Distance
51 FROM pgis_ravindranathan.lu_franchises fr
52 JOIN pgis_ravindranathan.fastfoods ff ON ff.franchise = fr.franchise_code,
53 pgis_ravindranathan.kingcounty
54 ORDER BY Distance ASC
55 LIMIT 3; -- LIMIT 3
56

```

Data Output Messages Notifications

Showing rows: 1 to 3

	franchise_name character varying (100)	distance double precision
1	McDonal's	260.28875412927
2	Burger King	266.2702786460578
3	Kentucky Fried Chicken	19398.802631640072

lygon,

)

Q7. Find how many fast-food franchises are within two miles of I-5.(1pt)

```

15
16 -- // Selecting specific franchises
17 SELECT fr.franchise_name, COUNT(DISTINCT ff.ff_id) AS tot
18 FROM pgis_ravindranathan.fastfoods AS ff
19 JOIN pgis_ravindranathan.lu_franchises AS fr ON ff.franchise = fr.franchise_co
20 JOIN pgis_ravindranathan.i5seattle_p AS i5 ON ST_DWITHIN(ff.geom, i5.geom, 1609
21 GROUP BY fr.franchise_name ORDER BY tot DESC;
22

```

Data Output Messages Notifications

Showing rows: 1 to 8

	franchise_name character varying (100)	tot bigint
1	McDonal's	22
2	Jack in the Box	14
3	Taco Bell	11
4	Burger King	10
5	Pizza Hut	8
6	Kentucky Fried Chicken	7
7	Wendy's	6
8	MOD Pizza	1

Total rows: 8 Query complete 00:00:00.457

Q8. Where is the closest franchise restaurant from the King County center point? (1pt)

Query Query History

```

1 SELECT fr.franchise_name,
2 ST_distance(ff.geom, ST_centroid(pgis_ravindranathan.kingcounty.geom)) AS Distance
3 FROM pgis_ravindranathan.lu_franchises fr
4 JOIN pgis_ravindranathan.fastfoods ff
5 ON ff.franchise = fr.franchise_code,
6 pgis_ravindranathan.kingcounty
7 ORDER BY Distance ASC
8 LIMIT 1;
9

```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1

	franchise_name character varying (100)	distance double precision
1	McDonal's	260.28875412927

I'm a little suspicious because this took 1 min 14s to finish, with only 4 new restaurants. Previously it was around 26 s

Q9. Provide two screen shots of the updated look up table and fast-food franchise location table. (1pt)

The new restaurants have franchise_code H, C, P, and M

Query Query History

```

1 SELECT * FROM pgis_ravindranathan.lu_franchises;
2

```

Data Output Messages Notifications

Showing rows: 1 to 14 Page No: 1

	franchise_code [PK] character (1)	franchise_name character varying (100)
1	b	Burger King
2	c	Carl's Jr
3	h	Hardee's
4	i	in-N-out
5	j	Jack in the Box
6	k	Kentucky Fried Chicken
7	m	McDonal's
8	p	Pizza Hut
9	t	Taco Bell
10	w	Wendy's
11	H	Habit Burger & Grill
12	C	Chipotle
13	P	Panda Express
14	M	MOD Pizza

Query Query History

1SELECT * FROM pgis_ravindranathan.fastfoods

2ORDER BY ff_id DESC;

3

Data OutputMessagesNotifications

Showing rows: 1 to 1000

Page No: 1

	franchise character (1)	lat double precision	lon double precision	ff_id [PK] integer	geom geometry
1	P	47.6374	-122.1408	50006	0101000020730800007B67E87F02EA38C1303184302ED81...
2	C	47.6186	-122.1834	50005	010100002073080000F6A8F8C828F838C19BC05EFE51C61...
3	H	47.6207	-122.03987	50004	01010000207308000053CE1AE9E4CF38C16622BF24C49B1...
4	M	47.6654	-122.3104	50003	0101000020730800001A5921B4481539C12B8F78FC3B212...
5	w	40.8902	-74.2567	50002	01010000207308000032A0A0A8C12F4041107BC712FA5CF...
6	w	41.3709	-74.6981	50001	0101000020730800005A34D77BAF993F41C6241DB2CB88F...
7	w	39.6321	-79.9568	50000	01010000207308000094405CDA32E73941EF895C0771C61...
8	w	32.2162	-80.7098	49999	010100002073080000BED61F0770933B416BD2D36E94843...
9	w	32.8789	-117.231	49998	0101000020730800008E6269F9617C38C1CE1543DB400B3...
10	w	37.4272	-81.5836	49997	01010000207308000003D715B3EF9E38416C958CB4C1382...
11	w	39.1526	-84.8773	49996	010100002073080000694B1BC98AC933413293117AE6362...
12	w	35.2606	-77.6498	49995	010100002073080000D767F4AD75A13E4102F9F04199D12...
13	w	38.0204	-84.9098	49994	010100002073080000C49D9410671234410A659762680624...
14	w	31.9274	-85.256	49993	01010000207308000042A08207F93D354170888991EA4C3...
15	w	38.1773	-87.5924	49992	010100002073080000A6BF1D43C0803041F0E4E8B85FAD2...
16	w	36.6298	-83.7044	49991	0101000020730800002CD418CD64133641D7DADC6286122...
17	w	38.1758	-84.8964	49990	0101000020730800001516A072800B34415C5CA5296A7F2...
18	w	35.1545	-79.4155	49989	0101000020730800006C9661781A503C41201A40C186692...
19	w	30.2853	-97.7419	49988	010100002073080000CBEAA4B6FBAE0A41F181DAC5B8DA...

Total rows: 50006Query complete 00:00:00.683

Q10. Provide a screen shot of all fastfood restaurants, restaurants within 2 miles, a 2 miles buffer from I-5, and I-5 highway in QGIS. The image should include a table tree showing your database and tables, layers, and features in a map view like the last image in Step 4. Drawing spatial data in QGIS. (2pts)

